

Polychlorinated dibenzo-p-dioxins and polychlorinated dibenzofurans (PCDD/PCDF) in food samples collected in southern Mississippi, USA.



In 1994, we analyzed 43 foodstuff samples from local supermarkets in southern Mississippi, USA, for PCDD/PCDF. 2,3,7,8-Cl₄DD could be quantified in 31 of these samples. On a lipid basis, levels in meat (0.53-1.10 pg I-TEQ/g) and dairy products (0.42-1.10 pg I-TEQ/g) were slightly lower than those reported from other industrialized countries. While levels in dairy samples from the United States and Europe are comparable, there is a difference in the contribution of individual congeners to the I-TEQ: for example, in milk samples from Germany approximately 40% of the I-TEQ is due to the presence of 2,3, 4,7,8-Cl₅DF while in the Mississippi samples this congener only contributes 16%. The highest concentrations of PCDD/PCDF in our study were detected in the farm-raised catfish (10.2-27.8 pg I-TEQ/g). A unique finding was that in addition to the 2,3,7,8-substituted PCDD/PCDF the catfish samples contained many non-2,3,7,8-substituted congeners. This is unusual because vertebrate animals selectively eliminate or metabolize the non-2,3,7,8-substituted congeners.